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REVIEW

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IoT and ML-based automatic irrigation system for smart agriculture system

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Abstract

The development of the Internet of Things (IoT) and machine learning (ML) technologies has triggered smart agricultural systems. In smart agriculture, irrigation management plays a major role to reduce water waste. The monitoring settings, hardware modules, communication technology, and storage systems used in smart irrigation systems were analyzed to determine the optimal nature of water flow. This assessment aims to give an overview of the current state of the irrigation system by taking into account weather and soil moisture. This paper provides a comprehensive review of the utilization of various hardware modules in smart irrigation systems. Moreover, various communication technologies that aid in data transfer for efficient smart irrigation are reviewed. The ML method used for prediction is also evaluated, as well as based storage technologies like the cloud and databases used to store data for predictive irrigation systems. As a result, the paper provides an overview of all the factors that contribute to irrigation systems' smart operation as well as potential future paths for improving agricultural systems through IoT.

1 | INTRODUCTION

The Internet of Things (IoT) is an emerging technology that has applications in precision agriculture. IoT-based farm automation can transform the agricultural domain from one a manual system to one that does not require human monitoring (Rawal, 2017). As water is a scarce resource and over wastage of such an essential resource should be minimized, water should be monitored for quantity and quality (García et al., 2020). To improve water efficiency, sensor-based monitoring equipment can be used for automated scheduling. To offer information on the complete crop water requirements, sensor data is often linked to environmental factors and the quality of the soil's water supply (Placidi et al., 2020). The irrigation system uses a variety of sensors, including soil moisture, temperature, and humidity sensors (Naik et al., 2018). Based on the sensed data hardware modules like Arduino, raspberry water irrigation is turned on and off (Rayes & Salam, 2019). The sensed data and estimated values are then stored in databases for future prediction (Kodali & Sahu, 2016).

Store data in a database or cloud requires wired communication like Bluetooth, Wi-Fi, Global System for Mobile communication (GSM), ZigBee, and so on. These technologies transfer the data from the hardware module to storage

Abbreviations: AIRA, agricultural irrigation recommendation and alert; ANN, artificial neural network; CSI, camera interface; CSIS, controlled subsurface irrigation system; ETo, evapotranspiration; EWMA, exponential weighted moving average; GPRS, general pocket radio services; GSM, global system for mobile communication; IoT, Internet of Things; KNN, *K*-nearest neighbor; ML, machine learning; MLP, multilayer perceptron; NRF, National Response Framework; PA, precision agriculture; PHP, hypertext processor; RBFN, radial basis function network; S-BIS, sensor-based irrigation scheduling; SMS, short message service; SSIM, structural similarity-based water valve management mechanism; UV, ultraviolet; WebGIS, web geographic Iiformation system; WSN, wireless sensor network.

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without a loss (Darshnam, 2015). The information acquired is then instantly saved on the web page and may be seen on a mobile device or a computer. The data from the field is sent to the cloud using cellular networks that are part of the worldwide system for mobile communication (Kumar et al., 2017). The deployment of enhanced machine learning (ML) models in smart irrigation systems can help farmers make better decisions (Abioye et al., 2022). According to Chlingaryan et al. (2018), ML serves as a powerful and flexible architecture for data-driven decision-making. Thus, IoT- and ML-based agricultural technology is utilized to reduce water usage and improve crop yields. The objective of the paper was to review the smart irrigation system. The content of the paper is organized as follows: Section 1 represents the introduction; Section 2 presents the literature survey of various approaches.

Core Ideas

- Irrigation management is crucial in smart agriculture for minimizing water waste.
- This apper presents the ideal type of water flow that was determined by smart irrigation systems.
- Monitoring configurations, hardware modules, communication technology, and storage systems were analyzed in the smart irrigation system.

2 | LITERATURE SURVEY

In recent years, numerous technologies and approaches have been utilized in irrigation systems, including monitoring parameters, hardware modules, communication technologies, and storage systems with ML algorithms for decision-making. These topics are covered in the following sections. Additionally, the following section describes the importance and restrictions of these strategies.

2.1 | Review on various monitoring parameters in smart irrigation

In smart irrigation, it is important to monitor weather and soil moisture conditions. Guravaiah and Raju (2020) proposed a wireless sensor-based networking system to reduce the impact of water and nutrients on yield. In the procedure of applying water to crops in this system, soil water sensors and atmospheric temperature and humidity sensors are used to start the irrigation. A rain sensor is utilized to monitor the weather, and if rain is forecast, all motors will be turned off. Second, liquid nutrients will be applied to the crop through the irrigation system. Data collection using River Formation Dynamics is used to gather and store the information on a server for later processing.

An intelligent irrigation system based on an algorithm that takes into consideration both observed data and forecasted weather for the following few days has been proposed by Velmurugan (2020). Web-based information visualization and decision support systems provide real-time information insights based on the analysis of sensor data and weather forecast data. Sensor node data was wirelessly acquired over the cloud utilizing web services. To check for leaf or plant discoloration and convey the results to cure the ailment from anywhere, the system must be manually managed and industrialized with video feeds.

Shekhar et al. (2017) developed an automated irrigation system in which sensor data on soil moisture and temperature is captured, and an KNN (*K*-nearest neighbor) classification ML algorithm was used to predict when to irrigate. This cost-effective system reduced water and labor requirements.

Keswani et al. (2019) used a self-contained IoT-enabled wireless sensor network (WSN) architecture to collect real-time farm data using multipoint measurements of soil moisture, soil temperature, ambient temperature, and environmental humidity. The anticipated observation strategy includes all freestanding IoT-enabled WSN nodes used for accurate data collection and archiving of agricultural information. The research explains how to use irrigation as effectively as possible by managing water valves precisely and predicting soil water needs using neural networks. This irrigation control method makes use of a structural similarity-based water valve management mechanism, which is used to detect agricultural regions with water shortages. The orders for valve control are processed using fuzzy logic-based weather condition modeling. Yet, it is important to take into account the human source needed to turn the valve on and off, as well as valve failure.

Adenugba et al. (2019) used sensors and environmental data from the Internet of Everything to manage and monitor a solar-powered smart irrigation system. Using the radial basis function network (RBFN), the acquired data was utilized to forecast environmental conditions (RBFN). The irrigation system is controlled using expected water level values, weather forecasts, humidity, temperature, and irrigation data. A web platform was created to provide remote monitoring and control of the system. When a precipitation event is forecast, the RBFN enables the conservation of natural resources by reducing applications of water.

Tyagi et al. (2017) developed a smart irrigation system based on monitoring the soil moisture to sense the moisture.

Many studies have created smart irrigation systems by monitoring soil water (Tyagi et al., 2017; Kiran Gowd et al.,

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20202; Ramachandran et al., 2018). In many of these systems, field data is data processor through the IoT.

In the work of Tiglao et al. (2020), the Wireless Sensor and Actuator Network technology has a mesh-like arrangement of in-field nodes that operate as a sensor for soil moisture, temperature, and humidity as well as an actuator on a drip irrigation valve. When changes in the network occur, the mesh-based network was dynamically structured to allow sensor nodes to self-reorganize. This system requires the development of a stable and long-lasting power supply for the in-field nodes, as well as the modification of the web application to display the nodes' power status. Additionally, the broadcast range should be expanded.

Dasgupta et al. (2019) recommended autonomous irrigation system monitoring to replace the manual system, which has issues with personal accountability and inexperienced field workers. This system keeps track of temperature, humidity, soil moisture content, and other physical factors. To evaluate if irrigation is necessary, the characteristics and crop yield were compared to information from earlier surveys. The system should concentrate on predicting weather features because previous surveys have never given trustworthy findings. Table 1 shows review on various monitoring parameters in smart irrigation.

Due to the cost of maintaining the solar system and valve failures, the current approaches are expensive. Furthermore, water seeps through the soil profile. A lot of water is wasted during Broadbent irrigation and rotating platform sprinkler irrigation. Moreover, irrigating a field with drip irrigation takes a while. The system then needed to change the web application so that it could display the nodes' power status as well as a dependable and long-lasting power supply for the in-field nodes. As a result, longer term irrigation based on weather forecasting has to be developed.

2.2 | Review on hardware modules used for smart irrigation

Fidaus et al. (2019) built a system that administers and monitors the irrigation system manually or automatically using the IoT in a WSN environment. This system used Arduino technology and the NRF24L01 as the communication channel's microcontroller and transceiver. It saves money on staff salaries and avoids water waste in everyday demands. Communication, on the other hand, should be enhanced.

Kumar et al. (2019) used the NodeMCU microcontroller to construct an IoT-based system for smart garden monitoring, which aids users in recognizing the current temperature, wetness, and humidity characteristics of their cultivated plants and gardens. This system provides a real-time profile of temperature, wetness, and humidity shown using an Android mobile application. This technology allows users to better manage plant health and improve problem-solving for gardens in large cities. To enhance system functionality, it should provide an interface for storing all previous data. Moreover, a warning against specific parametric values must be presented together with the basic care characteristics and precautions of plants.

Kanade et al. (2021) have created a project that uses IoT and ML to tackle these irrigation issues (ML). The hardware uses many sensors, including temperature, humidity, pH, and Arduino-controlled pressure sensors. This system can help users reduce their water use.

To monitor crop fields (Vineela et al. (2018), sensors for temperature, humidity, and soil moisture are recommended. By keeping an eye on these variables, the irrigation system may be automated if soil moisture levels are low. The automation system is connected to the remote location-controlling device using a Raspberry Pi 2. It boasts a quad-core 900 MHz ARM Cortex-A53 Processor and 1 GB of Memory. The board also features 40 GPIO pins, a Micro SD card slot, a Full HDMI connection, four USB ports, an Ethernet port, a 3.5 mm audio line, and video camera interface (CSI) and display interface connectors. The Raspberry Pi, however, cannot run on the Windows operating system, which leads to system overheating.

Silpa et al. (2018) used an Intel Galileo-based automated drip irrigation that takes advantage of the IoT benefits. This technology helps to decrease irrigation and labor. This method is cost and energy-efficient, and it can be used in small and large agricultural areas.

Suresh Kumar et al. (2021) proposed a framework that would improve water consumption based on soil moisture, climatic forecasts, and so on. It also uses IoT technology to alert its owner on the current state of the soil and irrigation systems—a dampness sensor to assess water stress. As a result of meeting the interest in water, the engine will shut down and switch back on when the field is dry. However, his approach is unreliable and unfit for use in an industrial setting.

Sunehra (2019) created a web-based smart irrigation system employing Raspberry Pi. The technology measures the soil's temperature and moisture content. When a 3-phase supply was provided in the field, the motor was automatically turned on and off based on water requirement. The NRF24L01 transceiver was used to send the temperature and moisture content of the soil to the Raspberry Pi. The received values are entered into the soil test blogs, the webpage displays pesticide and crop recommendations for the farmer to consider to enhance production. The device has the potential to cut water use and waste. Using HyperText Markup Language (HTML) and Hypertext Processor (PHP), a webpage was created to display the status of various field parameters, which can be accessed by anybody.

In Aminuddin et al. (2021), a smart irrigation system using Raspberry Pi has been proposed to ease irrigation. In this

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TABLE 1 Review on various monitoring parameters in smart irrigation

Reference	Techniques used	Significance	Problem identification
Guravaiah and Raju (2020)	Wireless sensor-based network with rain forecast data	• With rain forecasting data, it can save water • Crops get high nutrients by using a venturi injector	Large-scale implementation of this system is expensive
Velmurugan (2020)	Web services with weather prediction elements	Web-based information visualization and decision support systems give real-time information insights based on the analysis of sensor data and weather forecast data	The system will need to be operated manually, and it will also need to be industrialized with camera feeds to check for discoloration of leaves
Shekhar et al. (2017)	KNN classification machine learning algorithm based on captured temperature	• Reduced water consumption and human resources • Viable and cost-effective for maximizing agricultural water supplies	The water level in the tank should be considered before irrigating to field
Keswani et al. (2019)	SSIM-based irrigation system	SSIM identifies farm locations with water shortages	• Human sources are required to switch on/off the valve • Valve failures should be noted
Adenugba et al. (2019)	Radial basis function network	Avoiding unnecessary irritation when a precipitation event is predicted	The cost of maintaining a solar power system is expensive and surplus power generated by the solar system was wasted
Tyagi et al. (2017)	Monitoring soil moisture and using a microcontroller to turn ON/OFF the pump automatically	Hence helpful in saving water	Soil temperatures decrease water viscosity, thus allowing more water to percolate through the soil profile Broadbent
Kiran Gowd et al. (2020)	Moisture monitor and pump water with an alert message	• Low power consumption • On/off the system with a message alert	Rotating platform sprinkler makes lots of water waste, and this system is not suitable for all types of crops
Ramachandran et al. (2018)	Based on sensed soil moisture, pH, soil type, and weather conditions with optimized tools	Solenoid valves are used to automate the water flow rate, which saves water	• Voltage-sensitive and the coil may need to be changed at some point during the valve's lifespan. • The system requires improvements in optimization models and the development of an IoT prototype
Tiglao et al. (2020)	Wireless Sensor and Actuator Network-based a mesh-like configuration	• Act both as the sensor for soil moisture, temperature, and humidity and actuator on a valve • The drip irrigation method can save lots of water for future purposes	Drip irrigation takes a long time to irrigate the field
Dasgupta et al. (2019)	Physical monitoring of soil and compared with previous surveys	• Water flow initiated if only required	• Past surveys have never produced reliable results; thus, the system should focus on weather future prediction

Abbreviations: IoT, Internet of Thing; KNN, *K*-nearest neighbor; SSIM, structural similarity.

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system, three different sensors, including moisture, humidity, and temperature sensors, are installed in the soil of the plant. The collected data from the sensors is used to predict whether the plant needs to be watered or not. This system automates the irrigation by applying a logistic regression algorithm for predicting the irrigation. At the same time, the user monitors the status of the plant without having physical access to the plant through an Android application.

In the work of Tararani et al. (2018), Drip irrigation was designed as a smart irrigation technology to preserve water and nutrients for the crops. When the temperature, humidity, or current condition of rain in the surroundings changes, the surrounding sensors detect the change and send an interrupt signal to the Raspberry Pi. The goal of this system is to save time, money, labor, and energy.

Vij et al. (2020) designed an IoT-based irrigation system automation for farms, which would be a less expensive and more exact answer to farm demands. Their system is built on two types of microcontrollers Raspberry Pi 3 and the Arduino Mega. The changeable factors are continually checked with the use of various sensors, and irrigation was appropriate and suited to the type of crop. Extreme weather conditions, on the other hand, are a major concern. ML algorithms must be taught on enormous amounts of data as well as data from specific regions. For visualization, a dedicated server/network storage must be available. Table 2 shows review on hardware modules used for smart irrigation.

Various hardware modules such as Arduino Uno, Arduino mega, Node MCU, Raspberry Pi 2, Raspberry Pi 3, and Intel Galileo Gen-2 are reviewed, and the limitations are discussed in the table. Using those modules has produced overheating in the entire system and is not suitable for industrial operation. Moreover, the ML algorithm needs dedicated server/network storage for large data for displaying the data values, and the sensors' data to be graphically visualized. Moreover, communication between the node in extreme weather conditions needs improvement. Hence, further research should be carried out to perform even in 32-bit controllers with high Clock speed.

2.3 | Review on the communication technologies used in smart irrigation system

Srivastava et al. (2018) created a smart irrigation system with pH, water flow, temperature, and soil moisture sensors that detect pH, water flow, temperature, and moisture, and the Arduino microcontroller runs the servo motor and pumps depending on these sensors. Arduino received the data and wirelessly delivered it to the website through the internet using the ESP8266 Wi-Fi module. Using IoT, the transferred data was monitored and controlled. This allows the user to control the device remotely over a secure internet web connection. The website was created to operate water pumps and sprinklers to help farmers to increase output with higher quality crops. The ESP8266 Wi-Fi module, on the other hand, may easily allow a complete hijacking of an encrypted session.

Kavyashree and Shreedhara (2021) proposed Intelligent IoT-based Smart Irrigation System for an autonomous irrigation system that uses IoT technology and the KNN algorithm for ML. The smart irrigation system was designed to give appropriate water to crops while taking into account soil moisture and climatic conditions, preventing over- and under-irrigation without the need for human involvement. The GSM module was utilized in the system to convey the field's status to the farmer through short message service (SMS) and to record the parameters in the cloud. It minimizes over- and under-irrigation, reducing water waste and maintaining constant soil moisture in the field. The whole circuit was powered by electricity with a battery or other power source that can be used as a backup in the event of a power outage or damage. For more accurate and quicker predictions, advanced learner methods, such as decision trees, are required.

In the work of García et al. (2021), the impacts of the rural environment on the signal and the main variables to consider when developing a precision agriculture (PA) wireless network are investigated using various WSN deployment configurations for soil monitoring PA systems. The architecture, node design, and algorithm that determine watering requirements are all specified for the PA system. The testbed contains several types of vegetation as well as ESP32 Wi-Fi node placements on the ground, near the ground, and above the ground. The near-ground deployment provided the best coverage. Other regions monitored in PA systems, such as canals where irrigation water is conveyed, require additional research to discover the characteristics that impact communication in such environments.

In the work of Abayomi-Alli et al. (2018), the system utilizes wireless communication technology built utilizing the National Response Framework (NRF) module. The system is managed with ease using an Android app through Bluetooth. This system allows the user to choose between manual control for planned watering and automated control utilizing wireless sensors in the user experience. Water management was performed by the microcontroller–controlled solenoid valve operating on time. It may be used for planned watering both automatically and manually. However, the system must be improved by the use of smart mobile applications on the internet.

Gutiérrez et al. (2013) designed a system to maximize water consumption for crops, an automated irrigation system. A gateway device manages sensor data, initiates actuators, and sends data through General Pocket Radio Services (GPRS) to a web application. To manage water amounts, an algorithm-containing temperature and soil moisture threshold values were designed and implemented into a microcontroller-based

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TABLE 2 Review on hardware modules used for smart irrigation

Reference	Techniques used	Significance	Problem identification
Fidaus et al. (2019)	Wireless Sensor Network with Arduino and NRF24L01 microprocessor	Saves money on staff salaries and avoids water waste in everyday demands	Communication between the node to be improved
Kumar et al. (2019)	Smart garden monitoring using a NodeMCU microcontroller	Replaces the need for gardeners and issues faced during the maintenance of gardens in big cities	• System functionality should be improved • Overall maintenance properties and precautions of plants suggest against specific parametric values
Kanade and Prasad (2021)	Bolt IoT module and Arduino mega 2560	• Open source in hardware • Provide low-cost and low-power consumption	A controller can handle only 8 bits it does not support 32 bits, and the Clock speed is limited to 20 MHz
Vineela et al. (2018)	Connection automation with Raspberry Pi 2	• Support connecting various displays, modules, and sensors. • Fast processor and support all types of code like C, C++, java, and python	Raspberry Pi is not able to run on the Windows operating system, and it produces overheating issues to the whole system
Silpa et al. (2018)	Drip irrigation system using Intel Galileo	• Controls the water flow as well as the electrical flow • The system helps to save manpower, water, and time since irrigation is a time-consuming activity	• The system does not work automatically • Very expensive
Suresh Kumar et al. (2021)	Arduino uses the power of solar and wind energy	It works automatically as per the dirt dampness level. Subsequently, the wastage of water is reduced	This method is unreliable, and not suitable for industrial operation
Sunehra (2019)	Web-based smart irrigation system employing Raspberry Pi	The device has the potential to cut water use and waste	Using HTML and PHP, a webpage is created to display the status of various field parameters, which can be accessed by anybody
Aminuddin et al. (2021)	A smart irrigation system using Raspberry Pi with three different sensors	Allows users to monitor the status of the plant without having physical access to the plant through an Android application	The system instead of displaying the data values, the sensors' data to be graphically visualized over time
Tararani et al. (2018)	Drip irrigation using Raspberry Pi	Reducing the wastage of water and minimizing the manual labor on the field for irrigation to save time, cash, and power for the farmer	The system would provide a feedback control system that will monitor and control all the activities of plant growth and irrigation system efficiently
Vij et al. (2020)	Automation of irrigation systems with Raspberry Pi 3 and Arduino Mega	Use of various sensors, the variable parameters are constantly monitored, and irrigation is suitable for a specific type of crop	• Extreme weather conditions pose a serious threat • Machine learning algorithm needs to be trained on large as well as region-specific data • Dedicated server/network storage needs to be there for visualization purposes

Abbreviations: HTML, HyperText Markup Language; IoT, Internet of Thing; PHP, Hypertext Processor.

gateway. Photovoltaic panels powered the system, which included a duplex communication link based on a cellular–internet interface that allowed data examination and irrigation scheduling to be configured via a web page. The internet connection enables surveillance via mobile telecommunication devices such as a smartphone. However, when a large number of GPRS users in the same location use the GPRS services at the same time, the network might be damaged. Congestion occurs as a result of this, resulting in a slower data connection. In the event of a problem, troubleshooting is not feasible. This indicates that either GPRS services are operational or they are not.

Pravina et al. (2015) proposed an automated irrigation system based on controllable parameters such as temperature, soil moisture, and air humidity, which are critical aspects to be regulated in precision agriculture utilizing a WSN. The

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ZigBee technology is used to solve power-efficient time in WSNs. They employed ZigBee to accurate results for monitoring soil content, environmental content, and energy efficiency obtained. However, Its high maintenance costs, absence of a comprehensive solution, and tardy materialization, as well as its low transmission and network stability.

Ezhilazhahi and Bhuvaneswari (2017) created a remote monitoring system that continually checks the plant's soil moisture. The plant's soil moisture is continually monitored by the WSN, which is connected to the IoT. The Exponential Weighted Moving Average (EWMA) event detection technique was also utilized to extend the network lifespan. Data is also collected and sent to the server via ZigBee technology. The EWMA event detection method was utilized, and events are only generated when certain threshold conditions are satisfied. The nodes remain in a sleep state for most of the time, which allows them to conserve energy. It should be improved further by adding more low-cost sensor modules.

Savić and Radonjić (2018) designed the reception of the collected data on the main control unit, designed on the Waspmote microcontroller platform equipped with XBee S2 coordinator. The Waspmote platform is designed to communicate only with XBee routers and end devices. This architecture allows flexible implementation with a reliable data transfer that provides necessary information for autonomous decision-making in the smart irrigation system. However, the power consumption reduction of the system should be concentrated. Moreover, sensor nodes should be minimized.

In the study of Abhinaya and Sudhakar (2021), the irrigation system based on 6LoWPAN technology was introduced. Contiki OS and the Cooja simulator were used to simulate the 6LoWPAN network. Tmote Sky, which comes with Contiki OS, allows testing network output by simulating nodes. Smart irrigation systems are utilized to save water and energy while also allowing irrigation systems to be controlled and monitored remotely. Low personal area network (LWPAN) and gateway interoperability are improved via the 6LoWPAN network. The beacon interval duration for that node is also lowered when the radio service cycle is reduced, suggesting less power consumption and longer battery life. However, several networking technologies will be employed for network simulation.

The study of Kumar et al. (2018) includes soil wetness, temperature, and moisture sensor system that is deployed in the field. The framework was designed to find the proper field condition, manage water waste in the field, and provide accurate field control by employing the programed water system and the pieces that make up the basic equipment. Raspberry Pi small-scale controller, temperature sensor, mugginess sensor, soil dampness sensor, LCD, and GSM are all part of the irrigation framework. The GSM was used to communicate information to a control station or a computer, and as a result, the system is quite effective if we need to flood a large area. This framework is used to regulate and monitor the data that passes through the microcontroller. Although it has a restricted data rate capability, GSM advanced version devices are utilized for larger data rates. Table 3 shows review on the communication technologies used in smart irrigation system.

Various communication technologies used in smart automatic irrigation like Bluetooth, ZigBee, Wi-Fi, Ethernet, GSM, LoRa, and GPRS are reviewed, and the limitations of the technologies were given in the table. The range of communication of the system needs more study for other areas monitored in PA systems such as the canals where the irrigation water is transported to determine the factors that affect the communication in those environments. Moreover, the system needs to enhance over the internet using the advanced mobile application with increased flexibility, and connectivity costs were reduced. Hence further research should be carried out with some advanced technology for network simulation to communicate with nodes.

2.4 | Review on storage system used in smart irrigation system

Goap et al. (2018) presented an open-source technology-based smart system to predict the irrigation requirements of a field using the sensing of ground parameters like soil moisture, soil temperature, and environmental conditions along with the weather forecast data from the internet. The sensing nodes, involved in the ground and environmental sensing, consider soil moisture, soil temperature, air temperature, ultraviolet (UV) light radiation, and relative humidity of the crop field. The intelligence of this system is based on a smart algorithm, which considers sensed data along with the weather forecast parameters like precipitation, air temperature, humidity, and UV for the near future. The observed data is stored in the MySQL database. A responsive web-based user interface was developed using PHP, MySQL, and Bootstrap API for real-time monitoring and scheduling of irrigation activities. With the stored data in the database, the featured weather is predicted. The predicted value of the soil moisture is better in terms of its accuracy and error rate. However, the maintenance cost of the database is high, and water-saving analysis is to be monitored.

Rusdi et al. (2021) built a reporting system that is equipped with location tracking using the Smartphone Global Positioning System and the Web Geographic Information System (WebGIS) application. This field reporting system starts from the report input form on the Android app that detects location coordinates automatically. Reports are stored in an offline SQLite database and sent to the MySQL database server online through the data synchronization process. From the server side, information is displayed through the WebGIS application. Report flexibility through database

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TABLE 3 Review on the communication technologies used in smart irrigation system

Reference	Techniques used	Significance	Problem identification
Srivastava et al. (2018)	Information transmission with ESP8266 Wi-Fi module	• Enables the remote control mechanism through a secure internet web connection to the user • Keep an eye on the reference values which will help the farmer increase production with quality crops	ESP8266 Wi-Fi module can easily allow for a complete hijacking of the encrypted session
Kavyashree and Shreedhara (2021)	Intelligent IoT-based Smart Irrigation System with GSM-based communication	• The intelligent system eliminates much of the manual work that is stress-free for a farmer and mainly it avoids over- and under-irrigation state that in turn reduces the wastage of water and keeps constant soil moisture in the field • This system's action can be implemented with low power, low cost, small size, robust, and highly versatile	• The whole circuit works only on the power supply, in case of power cuts or damages a battery or other power unit can be implemented as an alternative source • KNN machine learning method is lazy learner it can be replaced with eager learner methods
García et al. (2021)	WSN deployment for PA systems	• PA system is described, providing the architecture, the node design, and the algorithm that determines the irrigation requirements • The best coverage obtained for the near-ground deployment	• The system needs more study for other areas monitored in PA systems such as the canals where the irrigation water is transported to determine the factors that affect the communication in those environments
Abayomi-Alli et al. (2018)	Wireless communication technology using NRF module	• User experience allows the selection of either manual control for scheduled irrigation or automatic control using wireless sensors • Water management is achieved by the timely operations of the solenoid valve controlled by the microcontroller	The system needs to enhance over the internet using an advanced mobile application
Gutiérrez et al. (2013)	Transmits data to a web application through GPRS	• This irrigation system can be adjusted to a variety of specific crop needs and requires minimum maintenance • The modular configuration of the automated irrigation system allows it to be scaled up for larger greenhouses or open fields • The internet link allows supervision through mobile telecommunication devices, such as a smartphone	• The network can be affected when more GPRS users in the same area utilize the GPRS services at the same time. This leads to congestion that results in slower data connection • It is not possible to troubleshoot in case of issues

(Continues)

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TABLE 3 (Continued)

Reference	Techniques used	Significance	Problem identification
Pravina et al. (2015)	Factors controlled in PA with ZigBee technology	• Accurate results for monitoring the soil content, environmental content, and efficient use of energy	• Its high maintenance cost, lack of total solution, and slow materialization • Low transmission, as well as low network stability. • Replacement with ZigBee-compliant appliances can be costly
Ezhilazhahi and Bhuvaneswari (2017)	EWMA event detection algorithm companies with ZigBee	EWMA event detection algorithm is used which generates events only when threshold conditions are met. The rest of the time, the nodes are in a sleep state which can save their energy	Further, it should be improved with more sensor modules at a low cost
Savić and Radonjić (2018)	Waspmote microcontroller platform equipped with XBee S2	Allows flexible implementation with a reliable data transfer that provides necessary information for autonomous decision-making in smart irrigation system	• Power consumption reduction of the system should be concentrated • Also, sensor nodes should be minimized
Abhinaya and Sudhakar (2021)	Irrigation system based on 6LoWPAN technology	• 6LoWPAN network improves LWPAN and gateway interoperability • When the radio service cycle is reduced, the beacon interval time for that node is reduced as well, indicating less power usage and a longer battery life	For network simulation, various networking technologies are to be used
Kumar et al. (2018)	Remote sensors with GSM communication	• The framework is exceptionally proficient on the off chance that we need to inundate a huge zone • Control and screen the information which is traversing the microcontroller	It provides limited data rate capability, for higher data rate GSM advanced version devices are used

Abbreviations: EWMA, exponential weighted moving average; GPRS, general pocket radio services; GSM, global system for mobile communication; IoT, Internet of Thing; KNN, *K*-nearest neighbor; LWPAN, low personal area network; NRF, National Response Framework; PA, precision agriculture; WSN, wireless sensor network.

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synchronization technology makes it easy for field workers to input reports online and offline. However, the SQLite system is lack multiuser capabilities.

Jariyayothin et al. (2018) focused on a soil-based plant with a water tank and tap, using the IoT and low-cost technologies. The system was divided into three parts: (1) a microcontroller system composed of NodeMCU V3 for sending and receiving sensor data, (2) an NoSQL database system that collects and stores monitoring sensor data as well as plant profiles, and (3) an IoT Backyard mobile application where users to choose or add plant profiles to crop, monitor water sensor data, and control watering sensors. It allows users to set up the plant for growth and create the plant profile to regulate soil moisture conditions and monitor current soil moisture and water levels in the tank. The Firebase data will be examined further to estimate plant growth rates and determine the optimal water level for each plant's kind and age.

Mustafa et al. (2021) designed a system that included regular checks of soil moisture content at the root zone and the start of the water system according to a premodified schedule based on the plant's concept, events, soil type, and environment. The sensors are set near the root zone in the ground for logical water system booking to determine the agent's moisture condition. A cloud-based and web-of-things-based water system framework utilized sensors to collect continuous water system data saves it in the cloud and then sends an order to the owner of the data, who then the user makes the appropriate action based on the conclusion. When the soil moisture level falls below a certain level, it automatically turns on. Sprinkler or water siphon failures, on the other hand, should be checked for adequate watering.

The work of Yamini and Reddy (2019) maintained the desired soil moisture level in the soil and sends the data to Arduino. Based on the sensed values, water will be supplied to the plant which can help to avoid over- and under-irrigation. A water sensor was used to sense the amount of water still available in the tank. Information from the two soil moisture sensors will be regularly updated on the web page using the Wi-Fi module and a message alert will be sent to the user. Sensor values are uploaded to the ThingSpeak channel to generate graphs for analysis, the same can be viewed in-app, and the motor can be controlled. This smart irrigation system is cost-effective, and it reduces the wastage of water. Real-time update to the cloud helps to view the current water condition in the plant. Continuous monitoring of parameters with various sensors increases power conception.

In the work of Ismail et al. (2019), the system developed with a few sensors such as (1) a soil moisture sensor (YL-69) to detect the water level in soil; (2) the humidity and temperature sensor (DHT-11) to trace early signs of temperature changes; and (3) the pressure sensor (BMP 280) to measure the pressure of the surrounding. These sensors are connected to a Wi-Fi module (Node MCU), and they are interdependent to give extra sensitivity to the irrigation system. The data collected will be uploaded to the cloud (ThingSpeak.com and Firebase) and displayed in the form of graphs that could be seen through the app and website. The app functions to display the reading from the sensors and to control the water pump for emergency purposes. This method reduces water consumption, minimal project cost, less labor, power consumption, and reliability. However, cloud storage is complex to develop and manage.

In the work of Mohammed et al. (2021), the ThingSpeak cloud platform was used to host sensor readings, do algorithmic analysis, show live data in real-time, deliver user event-based notifications, and transmit commands to IoT devices. The controlled subsurface irrigation system (CSIS) validation was used with sensor-based irrigation scheduling (S-BIS) to autonomously irrigate date palm plants is more efficient than using time-based irrigation scheduling. The S-BIS gave the date palm the right amount of irrigation water at the right time, right in the functional root zone. The developed CSIS's high efficiency boosted root development and soil nutrient absorption in the functional root zone. More research is needed to determine the influence of the current CSIS on various types of fruit trees and soil types.

Ramachandran et al. (2022) presented the development of irrigation and IoT, considerations for effective irrigation, the necessity for effective irrigation optimization, and the potential benefits of dynamic irrigation optimization for reducing water use. Additionally, various IoT architectural and deployment models, sensors, and controllers used in agriculture, IoT cloud platforms, well-known irrigation scheduling, and water need prediction tools and software by ML and neural network models for irrigation. In the future, ML and neural network approaches are used for effective analysis.

Elbeltagi et al. (2022) estimated the evapotranspiration (ETo) based on sparse meteorological data using an artificial neural network (ANN) technique. Reference crop ETo is a crucial part of the hydrological cycle that is utilized in other hydrological processes, such as irrigation, agricultural management, and water resource planning. The outcomes demonstrated many high-performance alternative approaches for estimating ETo in the absence of climatic data. The decision-making process for water managers, designers, and development planners can be aided by models that use ANN. However, the ANN approaches could be modified to assist policymakers and investors in future planning and development.

Phasinam et al. (2022) evaluated a smart irrigation system built on a cloud-based architecture and the IoT. To process this data in the cloud using a range of ML techniques, the system is built to assess soil moisture and humidity. Farmers receive accurate information regarding water content regulations. If farmers apply clever irrigation, they can reduce their water usage. In the future, to fill in all the gaps in the current design, a revolutionary IoT and cloud architectural idea need to be

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created. Table 4 shows the Review on storage system used in smart irrigation system.

Various storage systems are used to store and visualize the monitored data of the soil and weather conditions reviewed, and the significance and limitations of those storage systems are mentioned in the table. ThingSpeak cloud storage is used to analyze and predict plant growth rates and find appropriate water levels according to each plant type and age. However, this cloud storage cannot be accessed by any type of operating system platform. Moreover, these storage systems have some complexity when storing data. Hence, further research is required to develop efficient storage schemes to reduce the complexity and accessibility of multiple platforms. Moreover, an advanced analytical algorithm is required for future weather prediction.

2.5 | Review on various machine learning techniques in smart irrigation

Akshay and Ramesh (2020) introduced an advanced technology-based smart system to forecast the irrigation requirements of a field by sensing ground parameters such as soil moisture, temperature–humidity, and water level using an ML algorithm. They solved the overfitting problem using the KNN approach, which uses less memory and applies to all types of low-memory devices. However, it does not perform well with huge data sets as computing distances between each data instance is quite time-consuming, and it is vulnerable to noisy and missing data.

Veerachamy and Ramar (2022) built an Agricultural Irrigation Recommendation and Alert (AIRA) system that operates independently for each farm field register in the Hadoop Distributed File System. Water level alerts and other notifications are sent to all registered farm field owners. The information gathered will be analyzed using a hybrid classifier that combines an KNN and a neural network (k-N4). When the water level and pressure are low, alert messages are created based on real-time data from the farm fields. AIRA's recommendations will also help farmers make more money in the future. The AIRA system, on the other hand, should be extended with lightweight cryptography and tested by expanding the number of master nodes that can operate in various scenarios.

Bakthavatchalam et al. (2022) employed the supervised learning approach and ML algorithms to achieve the best model in the Waikato Environment for Knowledge Analysis. They chose the multilayer perceptron (MLP) rules-based classifier JRip and the decision table classifier as their ML algorithms for classification. Based on current environmental parameters, their smart module recommends the crop for watering and maximum production. Moreover, utilizing IoT, this ML algorithm assisted in the development of a model in the agriculture sector, assisting farmers in determining the best yield crop by simply measuring the required criteria. However, additional improvement is required in the smart agriculture system to minimize the drop in productivity, yield, or quality.

Priyanka et al. (2021) developed a distributed WSN, wherein each region of the farm is covered by various sensor modules which transmit data on a common server. ML algorithms were used to support predictions for irrigation patterns based on crops and weather scenarios. Moreover, the system establishes a properly distributed network that contributes to the accuracy of the predictions made by using a different algorithm. The communication speed of the system should be improved.

Raghuvanshi et al. (2022) provided a methodology for detecting and classifying intrusions into agricultural IoT networks. The NSL-KDD data set was used as an input data set in this system. All symbolic features in the NSL-KDD data set are first transformed into numeric features during preprocessing. Principal component analysis was used to extract features. To identify preprocessed data sets, ML methods such as support vector machine, linear regression, and random forest are utilized. This technology has improved efficiency, cost optimization, optimal energy use, forecasting, and convenience. However, the result of a prediction based on a certain data set is not always accurate.

Reddy et al. (2020) developed a machine-learning algorithm to predict the water requirement for a crop. This system comprises temperature, humidity, and moisture sensor, deployed in an agricultural field, sends data through a microprocessor, developing an IoT device with a cloud. They used a decision tree algorithm on the data sensed from the field to efficient results predict. The result from the decision tree algorithm is sent through a mail alert to the farmers to decide on the water supply. This system allows the former to start irrigation whenever required. Even though the system reduces water wastage it required human intervention. The system should have been developed with automatic irrigation.

In the work of Gomathy et al. (2021), ML prediction algorithms were used for predicting the temperature and humidity in the environment. This system uses various sensors and modules used to collect the data from the crop field. This method is worn to cultivate in areas where there is a shortage of water. In this irrigation system, the machine reads by contributing to the weather prediction. However, any single node failure affects the whole irrigation system; thus, it should be monitored.

Pavithra and Jayalakshmi (2020) developed a precision irrigation system to maintain the perfect water supply by using the random forest method, and ETo level wastage of water has been reduced. By using this technique, the system achieved better harvesting results and the total amount of water for lands and plants can be calculated by using the PH value. By using this method, proper utilization of water is achieved

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TABLE 4 Review on storage system used in smart irrigation system

Reference	Techniques used	Significance	Problem identification
Goap et al. (2018)	Open-source technology-based smart system	The predicted value of the soil moisture is better in terms of its accuracy and error rate	The maintenance cost of the database is high and water-saving analysis is to be monitored
Rusdi et al. (2021)	Field reporting using the Smartphone Global Positioning System	• Efficient in monitoring the performance of irrigation systems in real time • Report flexibility through database synchronization technology makes it easy for field workers to input reports online and offline	The SQLite system is lack multiuser capabilities
Jariyayothin et al. (2018)	Automated watering using the NoSQL database system	Enables users to set up the plant to grow and define the plant profile to control the soil moisture condition and monitor the current values of soil moisture and the water level in the tank	The data in the Firebase system will be further analyzed to predict plant growth rates and find appropriate water levels according to each plant type and age
Mustafa et al. (2021)	Consistent checking and storing of data in the cloud	Turned on automatically when soil moisture becomes low	Failures in sprinklers or water siphons should be monitored for proper irrigation
Yamini and Reddy (2019)	Real-time data stored in ThingSpeak cloud	• This smart irrigation system is cost-effective • The automatic watering system reduces the wastage of water • Real-time update to the cloud helps to view the current water condition in the plant	Continuous monitoring of parameters with various sensors increases power conception
Ismail et al. (2019)	ThingSpeak and Firebase cloud representation	• Displayed in the form of graphs that could be easy to analyze • The method reduces water consumption, minimal project cost, less labor, power consumption, and reliability	This cloud storage is complex to develop and manage
Mohammed et al. (2021)	ThingSpeak cloud platform as host sensor	Enhanced root growth and improved the soil nutrient	Further investigations are needed to study the impact of modern CSIS on different types of fruit trees and other soil types
Ramachandran et al. (2022)	Machine learning and neural network	Potential benefits of dynamic irrigation optimization for reducing water use	Water needs prediction tools for irrigation
Elbeltagi et al. (2022)	ANN	Estimate the ETo based on sparse meteorological data	The decision-making process for water irrigation is low
Phasinam et al. (2022)	Machine learning algorithm	Evaluated a smart irrigation system built on a cloud-based architecture and the Internet of Things	Reduce the water usage

Abbreviations: ANN, artificial neural network; ETo, evapotranspiration.

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based on the soil moisture level. However, to enhance agriculture production, the water level should be predicted with the recurrent neural network.

Togneri et al. (2019) built an architecture that enables multiple and customizable analytical approaches to precision irrigation, making room for the improvement of ML approaches. Impacts on different stakeholders were anticipated, including IoT professionals, by facilitating system deployment, and farmers, by providing cost reduction and safer crop yields. This system allows the farmer to highly customizable soil water management solutions, involving flexible connectivity among data, physical models, and ML algorithms. Moreover, this method is flexible data-driven soil water management, which allows suitable solutions to a great variety of soil, plant, and regional weather characteristics. However, the system needs to analyze the data and disseminate quantitative impact results.

Karar et al. (2020) used ML algorithms such as MLP neural network to support the decision of automatic control of IoT-based irrigation system to manage the water consumption effectively. This irrigation system supports the farmers to save wasted water, time, and effort to increase the productivity of their crops. It should study different crops for more effective use of water and also automatically measure the level of the water in the tank to alert the farmers via secured SMS mobile messages. Table 5 shows the review on various ML techniques in smart irrigation.

Various ML algorithms used in smart irrigation prediction such as KNN, random forest method, and MLP neural network are reviewed, and the limitations are given in the table. The prediction based on a certain data set is not always accurate as computing distances between each data instance is quite time-consuming and vulnerable to noisy and missing data. Moreover, the system should extend with lightweight cryptography and minimize the drop in productivity, yield, or quality. The further communication speed of the system should be improved to indicate the water level to the former via secured SMS mobile messages. Hence, further research should be carried out with some advanced algorithms to effectively control the use of water.

3 | RESULT AND SUMMARY

Various factors in the smart irrigation system were reviewed, and their results are depicted in this section.

- Irrigation based on monitoring tends to monitor the soil moisture and weather condition to reduce water consumption and human resources. However, continuous monitoring for irrigation reduces the moisture-holding capacity of the soil and increases erosion due to increased overland flow.
- The hardware monitoring system provides flexible irrigation to the crops. Raspberry Pi used in recent smart irrigation system support connecting various displays, modules, and sensors, and it can support a 32-bit controller. However, the Raspberry Pi module has issues like overheating and struggle to work in extreme weather conditions.
- LoRa communication technology in smart irrigation system communicates with nodes effectively and store data in the storage without loss. However, LoRa technology cannot continuously monitor weather conditions.
- ThingSpeak cloud storage stores the data for future prediction and for analysis of weather conditions and soil conditions which improves the production of crops with proper fertilizer to the field. However, the ThingSpeak cloud storage system cannot be accessible by all operating systems.
- The decision-making with an ML algorithm increases the flexibility of the irrigation system by using an KNN with a neural network. However, require a large data set for the decision-making method with the collection of real-time data from the farm fields.

4 | CONCLUSION

IoT and ML technologies make the irrigation process automatic with efficient usage of water by monitoring the soil and weather conditions. This paper reviewed the monitoring parameter, hardware modules, communication technology, storage, and ML algorithm for decision-making to improve the production of crops with optimum water supply, supply nutrients, and fertilizer to the crop. This analysis shows that various soil and weather parameters are used to reduce water usage. Further various hardware is used to reduce power consumption while monitoring. Various communication technology is used for communication to transmit data from sensor to storage with long-range using LoRa. Different storage displayed data in the form of graphs that could be easy to analyze. Then, the ML algorithm is used to make the irrigation decision for the future in the smart automatic irrigation system. Moreover, the future directions to overcome the issues in the existing automatic irrigation system are discussed.

5 | FUTURE PERSPECTIVE

In the future, the smart irrigation system can be enhanced by using the following methods and techniques:

- To improve production in a field with smart irrigation management, there is a need to monitor soil parameters based on weather conditions to avoid continuous irrigation.

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TABLE 5 Review on various machine learning (ML) techniques in smart irrigation

Reference	Techniques used	Significance	Problem identification
Akshay and Ramesh (2020)	*K*-nearest neighbor approach	Uses less memory and applies to all types of low-memory devices	• Does not perform well with huge data sets as computing distances between each data instance is quite time-consuming • Vulnerable to noisy and missing data
Veerachamy and Ramar (2022)	The *K*-nearest neighbor with a neural network (k-N4)	• Real-time data is collected from the farm fields, and alert messages are generated when the water level and pressure are low • AIRA recommendation improve profit in the farming field for the future	The system should extend with lightweight cryptography and test the system by increasing the number of master nodes that can act in different scenarios
Bakthavatchalam et al. (2022)	Supervised learning approach and ML algorithms	• Based on current environmental parameters, their smart module recommends the crop for watering and maximum production • Utilizing IoT, an ML algorithm assisted in the development of a model in the agriculture sector, assisting farmers in determining the best yield crop by simply measuring the required criteria	Additional improvement is required in smart agriculture systems to minimize the drop in productivity, yield, or quality
Priyanka et al. (2021)	Distributed wireless sensor network and ML algorithm	• Used to support predictions for irrigation patterns based on crops and weather scenarios • The system establishes a properly distributed network that contributes to the accuracy of the predictions	The communication speed of the system should be improved
Raghuvanshi et al. (2022)	Agricultural IoT networks with NSL-KDD data set	Technology has improved efficiency, cost optimization, optimal energy use, forecasting, and convenience	Prediction based on a certain data set is not always accurate
Reddy et al. (2020)	Decision tree algorithm	The system allows the former to start irrigation whenever required	The system should have been developed with automatic irrigation
Gomathy et al. (2021)	ML prediction algorithm	Worn to cultivate in areas where there is a shortage of water	One node failure affects the whole irrigation system; thus, it should be monitored
Pavithra and Jayalakshmi (2020)	Random forest method and evapotranspiration level	Proper utilization of water achieved based on the soil moisture level	To enhance agriculture production, the water level should be predicted with the recurrent neural network

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TABLE 5 (Continued)

Reference	Techniques used	Significance	Problem identification
Togneri et al. (2019)	IoT and ML	• The system allows the farmer to highly customizable soil water management solutions, involving flexible connectivity among data, physical models, and ML algorithms. • This method of flexible data-driven soil water management	The system needs to analyze the data and disseminate quantitative impact results
Karar et al. (2020)	MLP neural network	The system supports the farmers to save wasted water, time, and effort to increase the productivity of their crops	It should study different crops for more effective use of water and also automatically measure the level of the water in the tank to alert the farmers via secured SMS mobile messages

Abbreviations: AIRA, Agricultural Irrigation Recommendation and Alert; IoT, Internet of Thing; MLP, multilayer perceptron; SMS, short message service.

- Hardware usage cost and power usage as well as overheating should be minimized by using low-cost modules like Nano Every that is based on the ATMega4809 microcontroller with Wi-Fi and Bluetooth connectivity in it.
- The communication between sensors to store the data should be continuously monitored so that near-field communication technology can be used in the future that provides long-range communication and support for continuous monitoring.
- For effective modeling of the data storage system, Ubidots cloud can be used which can easily be accessed with the android system.
- The data analysis for future prediction requires advanced ML algorithms with density-based spatial clustering, which can discover clusters of different shapes and sizes from a large amount of data that contain noise.

AUTHOR CONTRIBUTIONS

Anoop E G: Conceptualization. **G. Josemin Bala**: Validation.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interests.

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